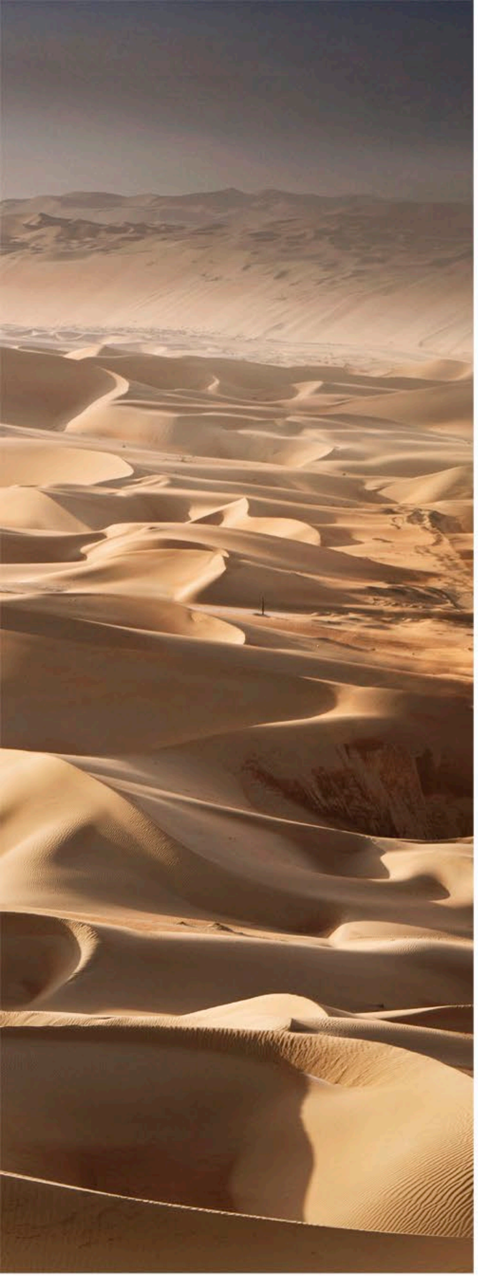
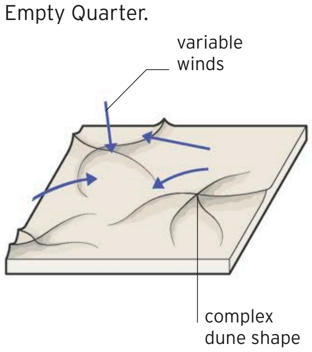


◁ **SHIFTING SEA OF SAND**
In the south of the desert lies Ar Rub 'al Khali, or the Empty Quarter, an area of sand roughly the size of France. Strong winds called *shamals* move huge amounts of sand twice a year, reshaping the dunes.

HOW A STAR DUNE FORMS

While most dunes are crescent-shaped, almost 10 percent of the planet's sand dunes are star-shaped. Winds blowing from several directions push piles of sand together, forming three or more "arms" of sand radiating from a high central point. The star dunes of the Arabian Desert are found in the eastern part of the Empty Quarter.



△ **LUNAR LANDSCAPE**
Southern Jordan's Wadi Rum, or the Valley of the Moon, reveals a volcanic past that created granite and sandstone mesas rising 2,620 ft (800 m) above the desert's surface.

Average rainfall in the Arabian Desert is less than **4 in (100 mm) a year**

The Dasht-e-Lut

A salt desert with spectacular dunes and one of the hottest places on the planet

Southeastern Iran's Dasht-e-Lut, or Lut Desert (*lut* means "completely barren"), is a torrid salt desert. In a satellite study of global land surface temperatures, the Lut's readings were the hottest for five out of seven years. In 2005, a temperature of 159.3°F (70.7°C) was recorded.

Black sand, high heat

The extreme heat is due partly to the desert's geology. Much of the Lut's surface is made up of black sand—mainly magnetite, a type of iron ore. Magnetite absorbs more radiant energy than light-colored sand, causing the surface temperatures to rise. Other distinguishing features of the Lut include yardangs. These huge, high rock ridges created by seasonal wind erosion give the desert a corrugated appearance, particularly on its western edge.

△ **MOBILE DUNES**
In the eastern part of the Lut's estimated 20,000 square miles (51,000 square km), desert winds shape the more sandy areas into massive mobile dune fields.